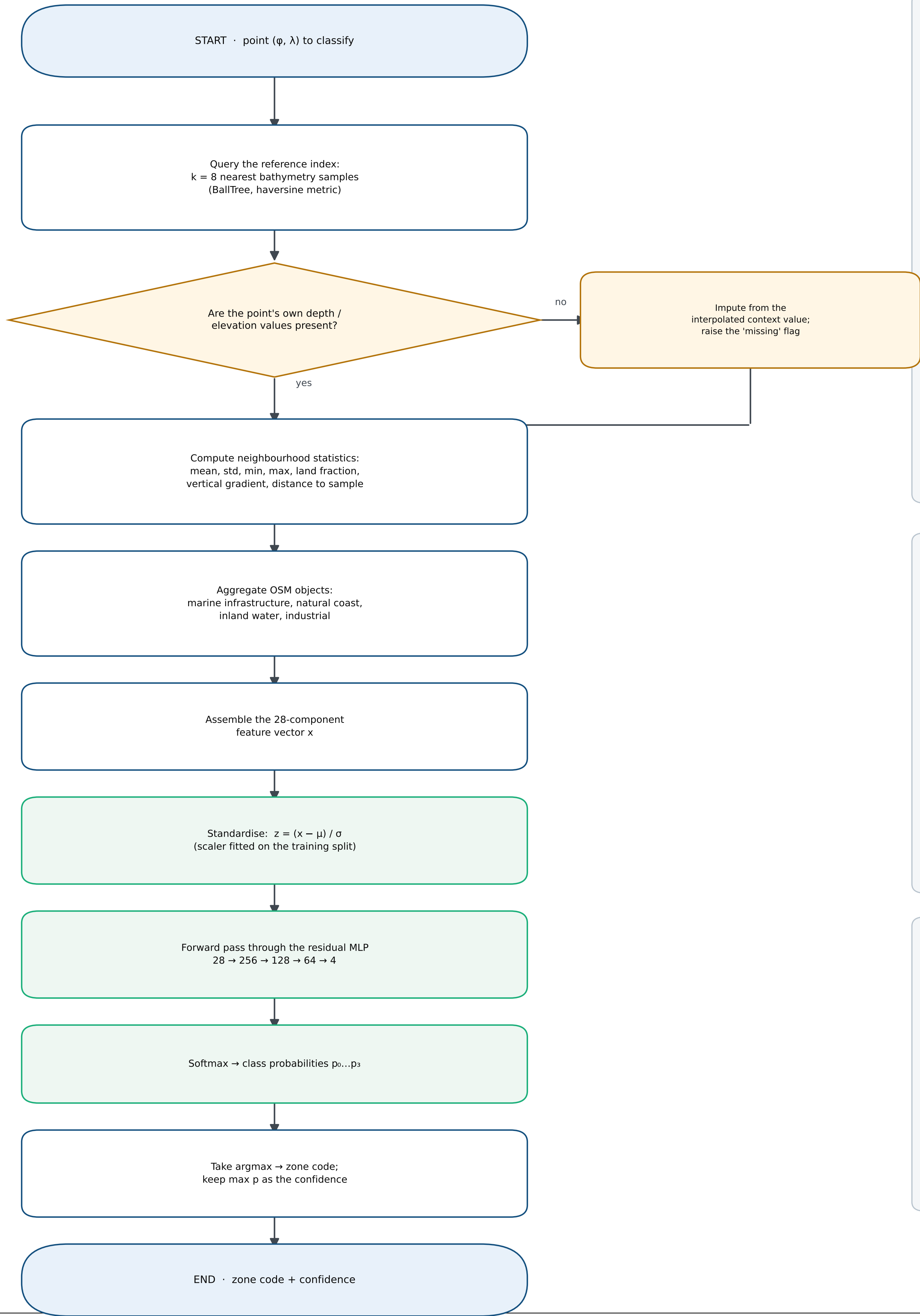




Feature vector (28 components)

Geodetic encoding — 4

$\sin \varphi, \cos \varphi, \sin \lambda, \cos \lambda$ (continuous across the antimeridian, no artificial discontinuity)

Point measurements — 5

signed-log depth, missing flag, height/depth flag, signed-log elevation, missing flag

Neighbourhood context — 9

mean / std / min / max depth of the 8 nearest samples, range, land fraction, sign agreement, log distance to the nearest sample, gradient

OSM objects — 10

marine infrastructure, natural coastline, inland water, industrial — each at the point and averaged over the neighbourhood; plus totals

Why a reference index is not leakage

- The index stores only depth, elevation and OSM counters — never a class label.
- It is built from the training split alone; validation and test points are never inserted.
- During training a point never queries itself (exclude_self = True).
- In deployment the same role is played by a NOAA bathymetry raster and an OSM extract, so the feature is reproducible for any coordinate on Earth.

Achieved quality

Class	Precision	Recall	F1	Support
OPEN_SEA	0.9766	0.9654	0.9709	8,205
COASTAL_SEA	0.9488	0.9603	0.9545	5,439
NEAR_COAST	0.9559	0.9770	0.9663	4,391
COASTLINE	0.9803	0.9639	0.9720	4,130
Overall accuracy			0.9662	22,165